GROUNDCONTROL
Global Satellite Communications

Making Ethernet Cables - Tricks of the Trade

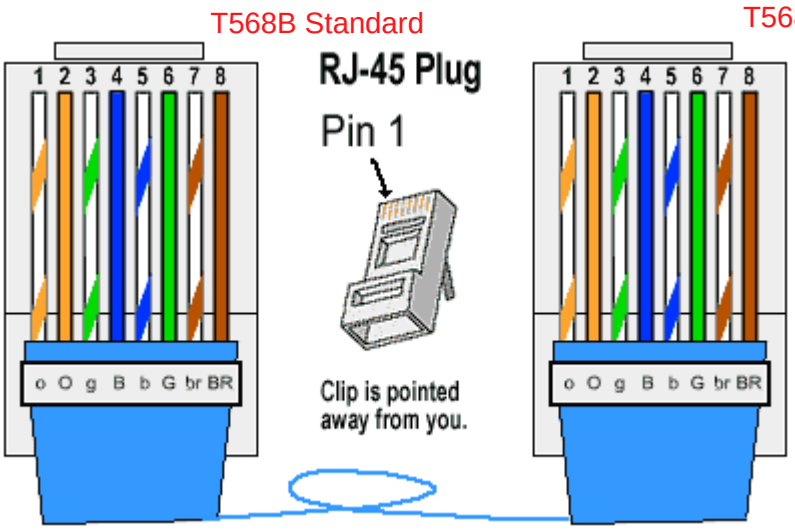
HOW TO MAKE AN ETHERNET CABLE
Purchasing Ethernet cables can be quite expensive and pre-made lengths are not always the length you need. Making Ethernet cables is easy with a box of bulk Category 5e Ethernet cable and RJ-45 connectors that are attached to the cut ends of your preferred cable length.

Bulk Ethernet Cable - Category 5e or CAT5e

(You may also use Category 6 or CAT6 cabling which has higher performance specifications and is about 20% more expensive than CAT5e.)

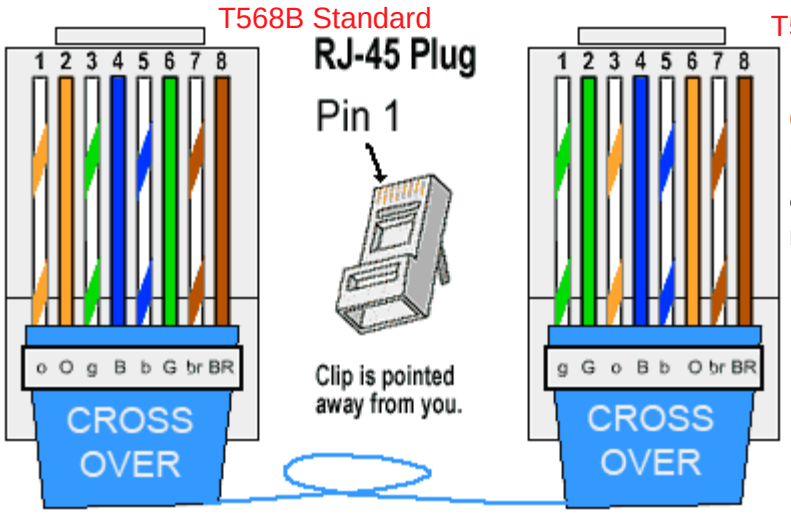
Bulk RJ45 Crimpable Connectors for CAT-5e
or
Bulk RJ45 Crimpable Connectors for CAT-6

RJ-45 Crimping tool

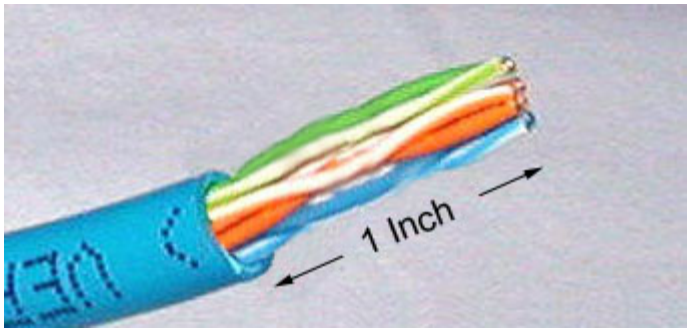


There are two kinds of Ethernet cables you can make, **Straight Through** and **Crossover**.

STRAIGHT THROUGH Ethernet cables are the standard cable used for almost all purposes, and are often called "patch cables". It is highly recommend you duplicate the color order as shown on the left. Note how the green pair is not side-by-side as are all the other pairs. This configuration allows for longer wire runs.

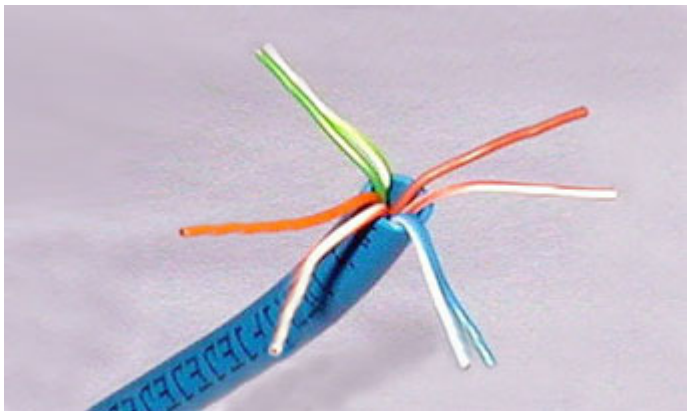


CROSSOVER CABLES - The purpose of a Crossover Ethernet cable is to directly connect one computer to another computer (or device) without going through a router, switch or hub.

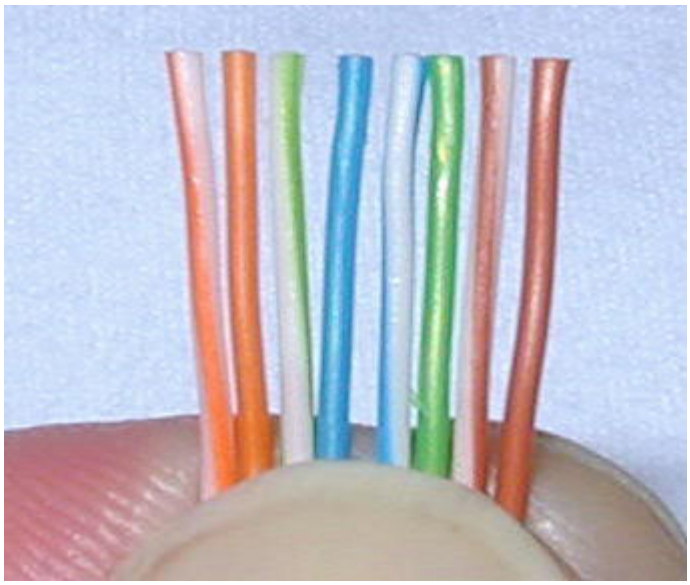


Here's how to make a standard cable:

cut into the plastic sheath about **1 inch** (2.5 cm) from the end of the cut cable. The crimping tool has a razor blade that will do the trick with practice.



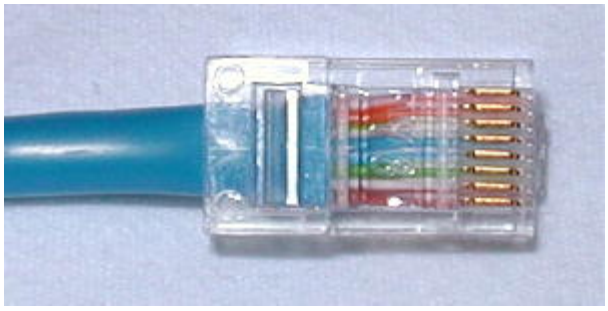
Unwind and pair the similar colors.



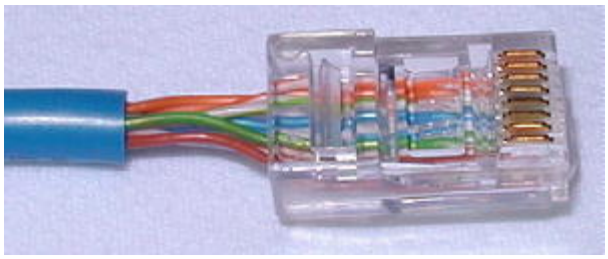
Pinch the wires between your fingers and straighten them out as shown. The color order is important to get correct.

Use scissors to make a straight cut across the 8 wires to shorten them to **1/2 Inch** (1.3 cm) from the cut sleeve to the end of the wires.

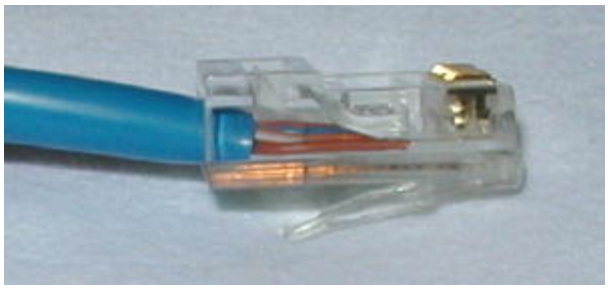
Carefully push all 8 colored wires into the connector. Note the position of the blue plastic sleeve. Also note how the wires go all the way to the end.



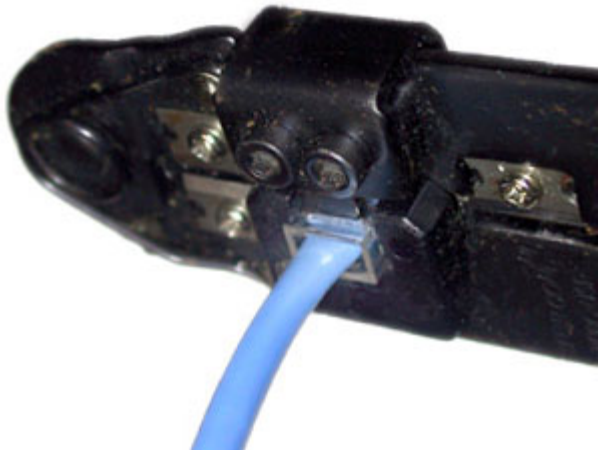
A view from the top. All the wires are all the way in. There are no short wires.



WRONG WAY - Note how the blue plastic sleeve is not inside the connector where it can be locked into place. The wires are too long. The wires should extend only 1/2 inch from the blue cut sleeve.



WRONG WAY - Note how the cables do not go all the way to the end of the connector.



CRIMPING THE CABLE ... carefully place the connector into the Ethernet Crimper and cinch down on the handles tightly. The copper splicing tabs on the connector will pierce into each of the eight wires. There is also a locking tab that holds the blue plastic sleeve in place for a tight compression fit. When you remove the cable from the crimper, that end is ready to use.

For a standard "Straight Through" cable, repeat all steps and wire color order on the other end of cable. For a cross-over cable, the other end will have a different color order as shown by the crossover picture above.

Make sure to test the cables before installing them. An inexpensive Ethernet cable tester does this quite well.

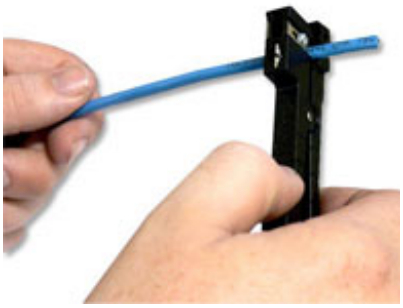
NOTE - The maximum cable length of CAT-5, CAT-5e or CAT-6 Ethernet cable is 328 feet or 100 meters.

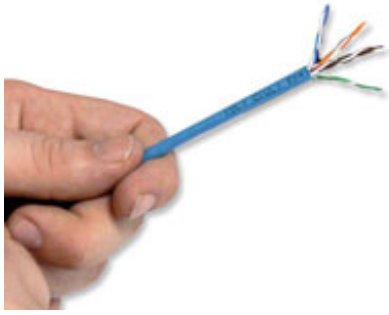
This .pdf file is a snapshot from http://www.groundcontrol.com/galileo/ch5-ethernet.htm

How to Terminate Punch Down Style Keystone Jacks

Due to numerous requests for wiring diagrams or general information on how to terminate cat5e and cat6 keystone jacks, ComputerCableStore™ has created the following "How to Terminate Punch Down Style Keystone Jacks (Cat5e / Cat6)". Within this "How To" article we have included everything you will need to find the materials, tools, and info on how to terminate Cat5e and Cat6 Keystone Jacks. We will cover materials needed, preferred tools and punch down technique.

Step 1: Make sure the stripper is adjusted properly on a scrap piece of cable. The Stripper should be adjusted to only score the jacket to make removing it easier and not nick the twisted pairs. Using a Coaxial Stripper strip around 1 inch of the jacket off. Be careful not to nick the conductors as this will cause the wire to break or short out the connection causing problems down the road. Inspect all wires for damage before proceeding to step 2.

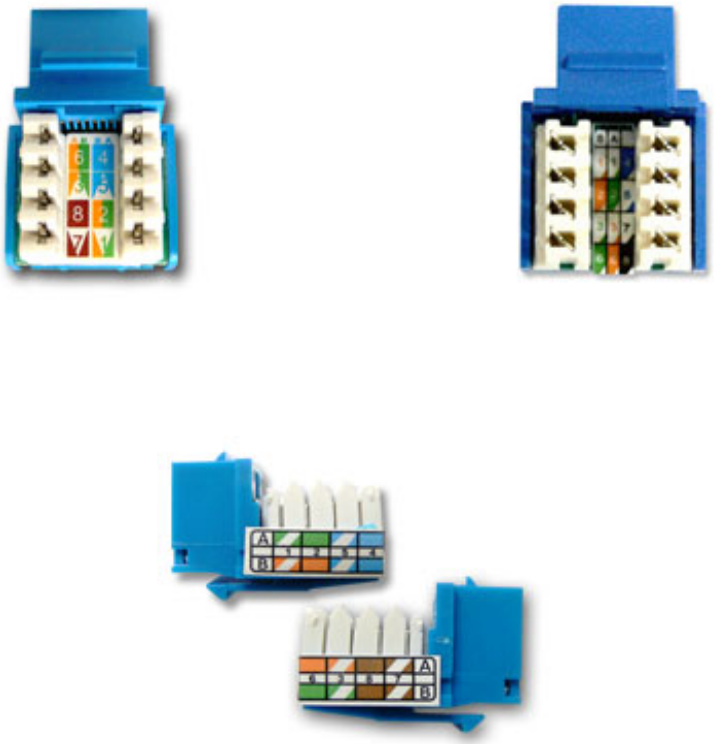




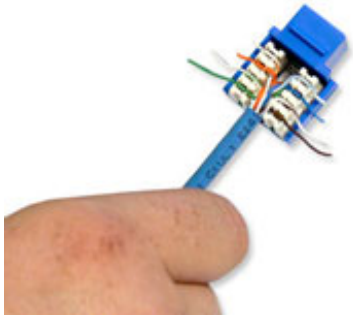
Step 2: Straighten the pairs out completely and lay them over the top of the keystone jack noting the color pattern for the 568b wiring. Note: Each keystone jack is slightly different in how they are labeled and how the colors are arranged. The 568B standard is most commonly used and ends of the cable must have the same standards to communicate. We have the 3 most

common keystone jack styles pictured here. The first jack pictured has the 2 standard pairs on the right, and the 2 variable pairs on the left. The A standard is the center column and the B standard is on the left. Both A and B standard applies to the right side of the jack. The solid color box with the lower right corner missing represents the solid color wire with the white stripe. The white box with the colored tip represents the white wire with the colored stripe.

The second jack pictured here has both A and B standards on both sides of the jack, with the color code running down the center. The solid color rectangle represents the solid color wire with the white stripe, wile the half white half color rectangle represents the white wire with the colored stripe. The third jack pictured here has the A and B standard codes labeled on the outside of the jack. With the A standard on top and the B standard on the bottom. The solid box represents the solid wire with the white stripe Wile the box with the white diagonal stripe going throught the middle, represents the white wire with the colored stripe.



Step 3: Keeping the pairs as twisted as possible press the wire into the correct groove with your thumbs. If you completely straighten the wires to run them through the jack you will risk cross talk between the pairs.



Step 4: Using a punch down tool punch the wires down into the blades built into the keystone jack. The blades in the jack are designed to work with solid cable, and may not work with stranded cable. Make sure the blade (as noted in the picture) is facing the outside of the keystone jack. If you reverse it you will cut the wires inside the keystone jack rendering them useless. The punch down tool should cut off the remaining pieces of the wire on the outside, but sometimes you may need to punch them down a 2nd time and wiggle the wire to it breaks clean off. Once this is done you can install the dust covers if your keystone jack comes with them. This keystone jack did not have the covers. The covers should never be used to do the job of the punch down tool.

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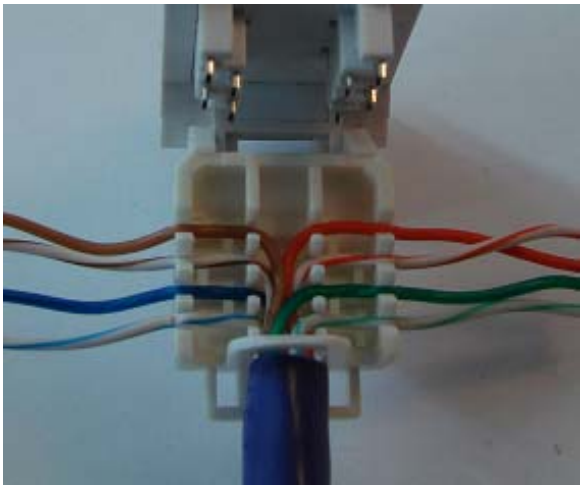
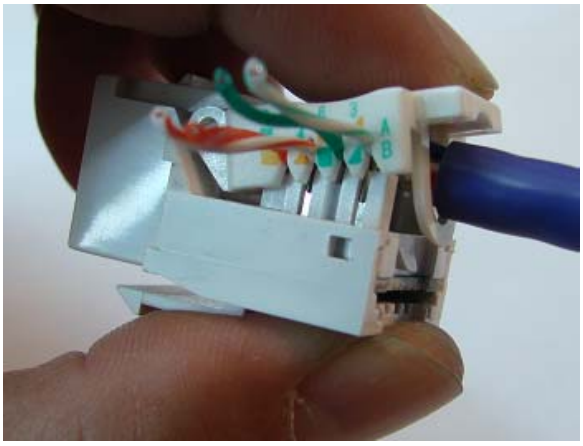
MONOPRICE
How to:
Connect a Cat5e/Cat6 RJ-45 Tooless Keystone Jack

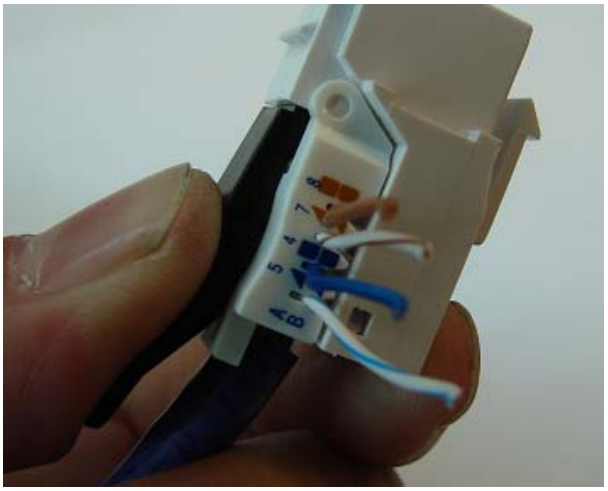
Step 1. Strip about 1” off of the outer jacket of the cable. If using Cat6 cable, trim off the spleen so that it doesn’t get in the way of the wires.



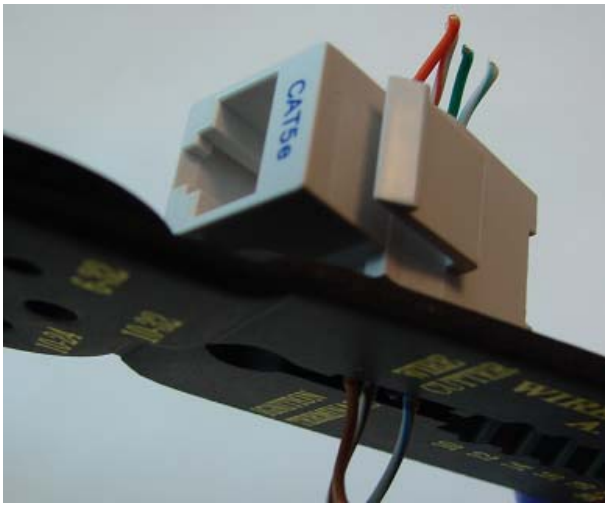
Step 2. Open the lid of the keystone and insert the wires through the loop at the end. Untwist and splay out the wires and sort them through the appropriate slots on the lid. Follow the color coding diagram on the side of the lid for the proper wiring standard you wish to use.



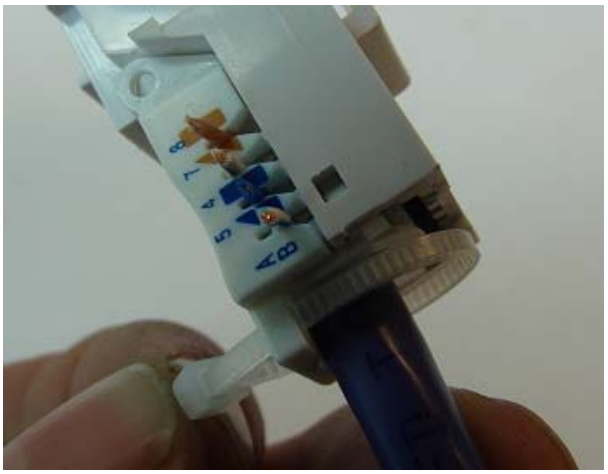
Step 3. Close the lid and squeeze tightly.



Step 4. Use the included lever to snap the lid all the way down.



Step 5. Trim away the excess wires from both sides of the keystone.



Step 6. If your model came with a zip tie, use it to secure the cable jacket to the keystone.